English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Software Engineering, Year 3, Groups 4/5, 2020/2021 Test 1

Your grade is 2 plus the number of points earned. Time allowed: 3 hours. Good luck.

Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. *Recall that, for* $A \subset \Omega$, *we write* $\overline{A} = \Omega \setminus A$.

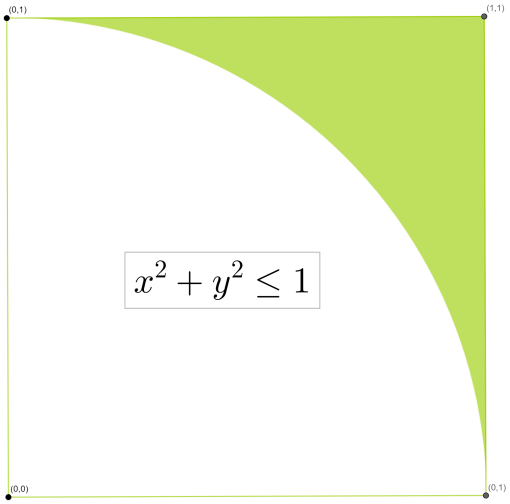
Problem 1. (1 point) Three dice are rolled together five times. What is the probability that the number of rolls showing only odd values is even? Find the expectation of this count.

• Let X be the number of rolls showing only odd values. Since the probability of all three dice showing odd values in one roll is $(1/2)^3$, we have $X \sim Bin(5, 1/8)$. Then

$$\begin{aligned}\mathbb{P}(X \text{ is even}) &= \mathbb{P}(X=0) + \mathbb{P}(X=2) + \mathbb{P}(X=4)\\ &= \binom{5}{0}\left(\frac{1}{8}\right)^0\left(\frac{7}{8}\right)^5 + \binom{5}{2}\left(\frac{1}{8}\right)^2\left(\frac{7}{8}\right)^3 + \binom{5}{4}\left(\frac{1}{8}\right)^4\left(\frac{7}{8}\right)^1\\ &\approx 0.619\,.\end{aligned}$$

By the standard formulas, $\mathbb{E}X = 5 \cdot 1/8 = 0.625$ and $DX = 5 \cdot 1/8 \cdot 7/8 \approx 0.547$.

Problem 2. (0.8 points) Two numbers are chosen at random from $[0, 1]$. What is the probability that the sum of their squares exceeds 1?



Answer: $1 - \pi/4 \approx 0.215$.

Problem 3. (1 point) A race has 25 teams: 8 in the jeep category, 10 in trucks and 7 in motorcycles. Jeeps finish with probability 0.9, trucks with probability 0.7, and motorcycles with probability 0.6. After the race, three teams are chosen randomly from the participants. It is known that one of the selected teams finished and the other two did not. What is the probability that the three selected teams belong to different categories?

• Let $A = \{\text{one selected team finished and the other two did not}\}$. Write K for truck, J for jeep and M for motorcycle. Define the hypotheses $H_{ijk} = \{\text{the selected categories are } \{i, j, k\}\}$ (there are $\binom{3+3-1}{2} = 10$ of them; why?). In this notation we seek $\mathbb{P}(H_{KJP}|A)$, knowing

$$\mathbb{P}(A|H_{KKK}) = 3 \cdot (0.3)^2 \cdot (0.7) = 0.189; \qquad \mathbb{P}(H_{KKK}) = \frac{\binom{10}{3}}{\binom{25}{3}} = \frac{6}{115};$$

$$\mathbb{P}(A|H_{JJJ}) = 3 \cdot (0.1)^2 \cdot (0.9) = 0.027; \qquad \mathbb{P}(H_{JJJ}) = \frac{\binom{8}{3}}{\binom{25}{3}} = \frac{14}{575};$$

$$\mathbb{P}(A|H_{MMM}) = 3 \cdot (0.4)^2 \cdot (0.6) = 0.288; \qquad \mathbb{P}(H_{MMM}) = \frac{\binom{7}{3}}{\binom{25}{3}} = \frac{7}{460};$$

$$\mathbb{P}(A|H_{KKJ}) = 2 \cdot (0.7) \cdot (0.3) \cdot (0.1) + (0.3)^2 \cdot (0.9) = 0.123; \qquad \mathbb{P}(H_{KKJ}) = \frac{\binom{10}{2}\binom{8}{1}}{\binom{25}{3}} = \frac{18}{115};$$

$$\mathbb{P}(A|H_{KKM}) = 2 \cdot (0.7) \cdot (0.3) \cdot (0.4) + (0.3)^2 \cdot (0.6) = 0.222; \qquad \mathbb{P}(H_{KKM}) = \frac{\binom{10}{2}\binom{7}{1}}{\binom{25}{3}} = \frac{63}{460};$$

$$\mathbb{P}(A|H_{JJK}) = 2 \cdot (0.9) \cdot (0.1) \cdot (0.3) + (0.1)^2 \cdot (0.7) = 0.061; \qquad \mathbb{P}(H_{JJK}) = \frac{\binom{8}{2}\binom{10}{1}}{\binom{25}{3}} = \frac{14}{115};$$

$$\mathbb{P}(A|H_{JJM}) = 2\cdot(0.9)\cdot(0.1)\cdot(0.4) + (0.1)^2\cdot(0.6) = 0.078; \qquad \mathbb{P}(H_{JJM}) = \frac{\binom{8}{2}\binom{7}{1}}{\binom{25}{3}} = \frac{49}{575};$$

$$\mathbb{P}(A|H_{MMK}) = 2\cdot(0.6)\cdot(0.4)\cdot(0.3) + (0.4)^2\cdot(0.7) = 0.256; \qquad \mathbb{P}(H_{MMK}) = \frac{\binom{7}{2}\binom{10}{1}}{\binom{25}{3}} = \frac{21}{230};$$

$$\mathbb{P}(A|H_{MMJ}) = 2\cdot(0.6)\cdot(0.4)\cdot(0.1) + (0.4)^2\cdot(0.9) = 0.192; \qquad \mathbb{P}(H_{MMJ}) = \frac{\binom{7}{2}\binom{8}{1}}{\binom{25}{3}} = \frac{42}{575};$$

$$\mathbb{P}(A|H_{KJM}) = (0.7)\cdot(0.1)\cdot(0.4) + (0.3)\cdot(0.9)\cdot(0.4) + (0.3)\cdot(0.1)\cdot(0.6) = 0.154; \quad \mathbb{P}(H_{KJM}) = \frac{\binom{10}{1}\binom{8}{1}\binom{7}{1}}{\binom{25}{3}} = \frac{28}{115}.$$

By Bayes' formula, substituting the values found above, we obtain

$$\mathbb{P}(H_{KJP}|A) = \frac{\mathbb{P}(A|H_{KJM})\mathbb{P}(H_{KJM})}{\sum_{i,j,k}\mathbb{P}(A|H_{ijk})\mathbb{P}(H_{ijk})} = \frac{0.154\cdot\frac{28}{115}}{\sum_{i,j,k}\mathbb{P}(A|H_{ijk})\mathbb{P}(H_{ijk})} \approx 0.244\,.$$

Problem 4. (1.2 points)

1. (0.4 points) Let $A, B \subset \Omega$ be events. Recall the definition of independence of A and B. Prove that if A and B are independent, then $\overline{A}$ and B are independent too.

   • A and B are independent if $\mathbb{P}(A\cap B) = \mathbb{P}(A)\mathbb{P}(B)$. If this holds, since $B = (A\cap B)\cup(\overline{A}\cap B)$, we have

   $$\mathbb{P}(\overline{A}\cap B) = \mathbb{P}(B) - \mathbb{P}(A\cap B) = \mathbb{P}(B)(1-\mathbb{P}(A)) = \mathbb{P}(B)\mathbb{P}(\overline{A}),,$$

   which proves the claim.

2. (0.4 points) Let $n > 1$ be an integer and $A_1, \dots, A_n$ events. Recall that $A_1, \dots, A_n$ are mutually independent if, for every $\{i_1, \dots, i_k\} \subset \{1, \dots, n\}$,

   $$\mathbb{P}(A_{i_1})\mathbb{P}(A_{i_2})\dots\mathbb{P}(A_{i_k}) = \mathbb{P}(A_{i_1}\cap A_{i_2}\cap\dots\cap A_{i_k})\,. \tag{1}$$

   Prove that if $A_1, \dots, A_n$ are mutually independent, then $\overline{A_1}, \dots, \overline{A_n}$ are also mutually independent.

   • One can use induction or inclusion–exclusion. Here is another proof: we show that if $A_1, \dots, A_n$ are mutually independent, then so are $A_1, \dots, A_{n-1}, \overline{A_n}$. The requested result follows by applying this n times. We verify definition (1) for the sets $A_1, \dots, A_{n-1}, \overline{A_n}$. Choose $I := \{i_1, \dots, i_k\} \subset \{1, \dots, n\}$. If $n \notin I$, then (1) follows from the independence of $A_1, \dots, A_{n-1}$. Otherwise, without loss of generality, $i_1 = n$. Then

   $$\begin{aligned}\mathbb{P}(\overline{A_n}\cap A_{i_2}\cap\dots\cap A_{i_k}) &= \mathbb{P}(A_{i_2}\cap\dots\cap A_{i_k}) - \mathbb{P}(A_n\cap A_{i_2}\cap\dots\cap A_{i_k})\\ &= \mathbb{P}(A_{i_2}\cap\dots\cap A_{i_k})(1-\mathbb{P}(A_n))\\ &= \mathbb{P}(A_{i_2})\dots\mathbb{P}(A_{i_k})\mathbb{P}(\overline{A_n})\,,\end{aligned}$$

   as required.

3. (0.4 points) Let X take values in $\mathbb{N}$, and for $n \in \mathbb{N}$ put $p_n := \mathbb{P}(X = n)$. Define the expectation $\mathbb{E}X$ and variance DX. Prove that

   $$\sum_{k=1}^{\infty}\mathbb{P}(X \ge k) = \mathbb{E}X\,. \tag{2}$$

•

$$\mathbb{E}X = \sum_{n=1}^{\infty} np_n; \quad DX = \mathbb{E}(X-\mathbb{E}X)^2 = \sum_{n=1}^{\infty}(n-\mathbb{E}X)^2 p_n\,.$$

For a formal proof of (2), interchange the order of summation, using the discrete form of Fubini's theorem:

$$\mathbb{E}X = \sum_{n=1}^{\infty} np_n = \sum_{n=1}^{\infty}\sum_{k=1}^{n} p_n = \sum_{k=1}^{\infty}\sum_{n=k}^{\infty} p_n = \sum_{k=1}^{\infty}\mathbb{P}(X\ge k)\,.$$

This formalizes the idea you were meant to notice:

$$\begin{aligned}\mathbb{P}(X\ge 1) &= p_1 + p_2 + p_3 + p_4 + \dots\\ \mathbb{P}(X\ge 2) &= p_2 + p_3 + p_4 + \dots\\ \mathbb{P}(X\ge 3) &= p_3 + p_4 + \dots\\ \mathbb{P}(X\ge 4) &= p_4 + \dots\,,\end{aligned}$$

which explains why $\mathbb{P}(X\ge 1) + \mathbb{P}(X\ge 2) + \dots = p_1 + 2p_2 + 3p_3 + \dots = \mathbb{E}X$. If desired, verify the argument as an exercise in discrete structures.